

Question **1**

Correct

Marked out of
3.00

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Given an array A of sorted integers and another non negative integer k , find if there exists 2 indices i and j such that $A[i] - A[j] = k$, $i \neq j$.

Input Format

1. First line is number of test cases T . Following T lines contain:
2. N , followed by N integers of the array
3. The non-negative integer k

```

1  #include <stdio.h>
2  int main()
3  {
4      int t;
5      scanf("%d", &t);
6      while(t--)
7      {
8          int n;
9          scanf("%d", &n);
10         int a[n];
11         for(int i=0; i<n; i++)
12         {
13             scanf("%d", &a[i]);
14         }
15         int k;
16         scanf("%d", &k);
17         int flag = 0;
18         for(int i = 0; i<n; i++)
19         {
20             for(int j = i+1; j<n; j++)
21             {
22                 if(a[i]-a[j]==k || a[j]-a[i]==k)
23                 {
24                     flag = 1;
25                     break;
26                 }
27             }
28
29             if (flag) break;}
30         printf("%d\n", flag);
31     }
32 }
33 return 0;
34 }

```

	Input	Expected	Got	
✓	1 3 1 3 5 4	1	1	✓
✓	1 3 1 3 5 99	0	0	✓

Passed all tests! ✓

Question 2

Correct

Marked out of
5.00

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Sam loves chocolates and starts buying them on the 1st day of the year. Each day of the year, x , is numbered from 1 to Y . On days when x is odd, Sam will buy x chocolates; on days when x is even, Sam will not purchase any chocolates.

Complete the code in the editor so that for each day N_i (where $1 \leq x \leq N \leq Y$) in array `arr`, the number of chocolates Sam purchased (during days 1 through N) is printed on a new line. This is a function-only challenge, so input is handled for you by the locked stub code in the editor.

Input Format

The program takes an array of integers as a parameter.

The locked code in the editor handles reading the following input from `stdin`, assembling it into an array of integers (`arr`), and calling `calculate(arr)`.

The first line of input contains an integer, T (the number of test cases). Each line i of the T subsequent lines describes the i th test case as an integer, N_i (the number of days).

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int t;
6     scanf("%d", &t);
7     while(t-->0)
8     {
9         int n,c=0;
10        scanf("%d", &n);
11        for(int i=0 ; i<=n; i++)
12        {
13            if(i%2!=0) c=c+i;
14        }
15        printf("%d\n", c);
16    }
17    return 0;
18 }
19 }
```

	Input	Expected	Got	
✓	3 1 2 3	1 1 4	1 1 4	✓
✓	10 71 100 86 54 40 9 77 9 13 98	1296 2500 1849 729 400 25 1521 25 49 2401	1296 2500 1849 729 400 25 1521 25 49 2401	✓

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

The number of goals achieved by two football teams in matches in a league is given in the form of two lists. Consider:

- Football team A, has played three matches, and has scored { 1 , 2 , 3 } goals in each match respectively.
- Football team B, has played two matches, and has scored { 2 , 4 } goals in each match respectively.
- Your task is to compute, for each match of team B, the total number of matches of team A, where team A has scored less than or equal to the number of goals scored by team B in that match.
- In the above case:
- For 2 goals scored by team B in its first match, team A has 2 matches with scores 1 and 2.
- For 4 goals scored by team B in its second match, team A has 3 matches with scores 1, 2 and 3.

Hence, the answer: {2, 3}.

Complete the code in the editor below. The program must return an array of m positive integers, one for each $maxes[i]$ representing the total number of elements $nums[j]$ satisfying $nums[j] \leq maxes[i]$ where $0 \leq j < n$ and $0 \leq i < m$, in the given order.

It has the following:

`nums[nums[0],...nums[n-1]]`: first array of positive integers

`maxes[maxes[0],...maxes[m-1]]`: second array of positive integers

```

1 #include <stdio.h>
2 int main()
3 {
4     int a,b,c;
5     scanf("%d", &a);
6     int ta[a];
7     for(int i=0; i<a; i++)
8         scanf("%d", &ta[i]);
9     scanf("%d", &b);
10    int tb[b];
11    for(int i=0; i<b; i++)
12        scanf("%d", &tb[i]);
13    for(int i=0; i<b; i++)
14    {
15        c=0;
16        for(int j=0; j<a; j++)
17        {
18            if(tb[i]>=ta[j]) c++;
19        }printf("%d\n", c);
20    }
21    return 0;
22 }

```

	Input	Expected	Got	
✓	4	2	2	✓
	1	4	4	
	4			
	2			
	4			
	2			
	3			
	5			
✓	5	1	1	✓
	2	0	0	
	10	3	3	
	5	4	4	
	4			
	8			
	4			
	3			
	1			
	7			
	8			